



IP Law meets Open Source Hardware

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Introduction: Research Focus

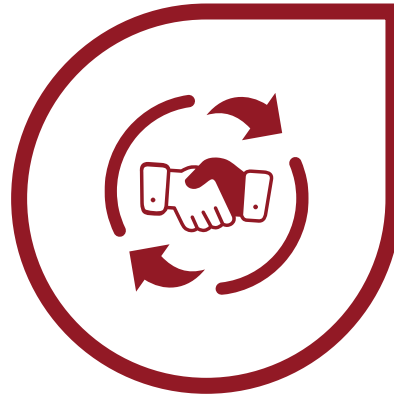
Creation of guidelines



IP-Rights and Circular Economy



Open Source Hardware
licensing models



Technology-sensitive liability
law in the context of Open
Source Hardware

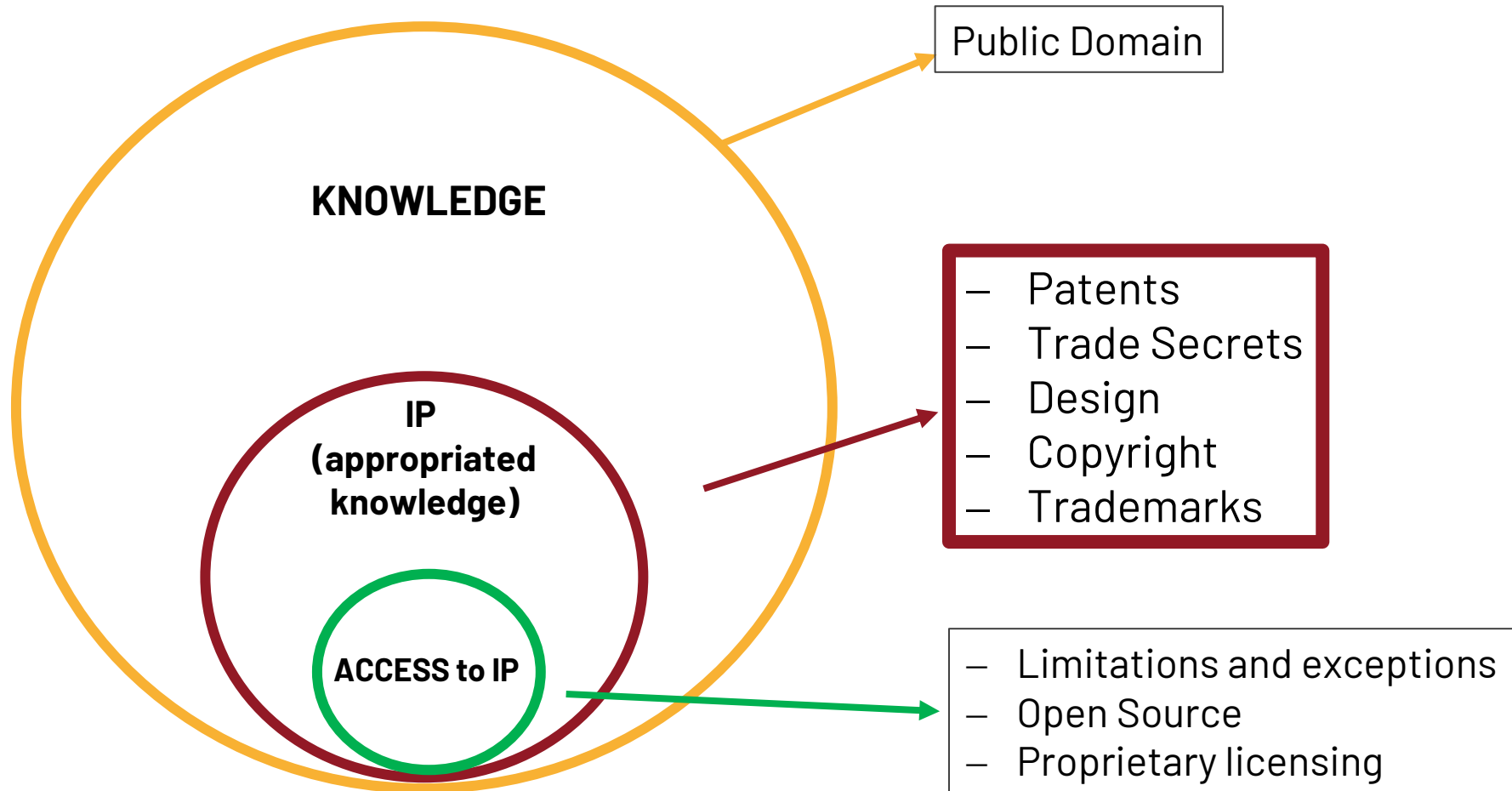


Agenda

- 1 Relevance of IP – Overview
- 2 IP Rights
- 3 Licensing – Overview
- 4 Case Studies

1. Relevance of IP – Overview

Why does IP exist at all?



IP Benefits

Reputation

Negotiate without
loss of protection

Provide Freedom
To Operate (FTO)

Licensing to
maintain control

Incentive to
foster innovation

Share information
with partners
= knowledge
distribution

Commercial
Interests and
economic
growth

Protection against
Infringement
and
fair competition

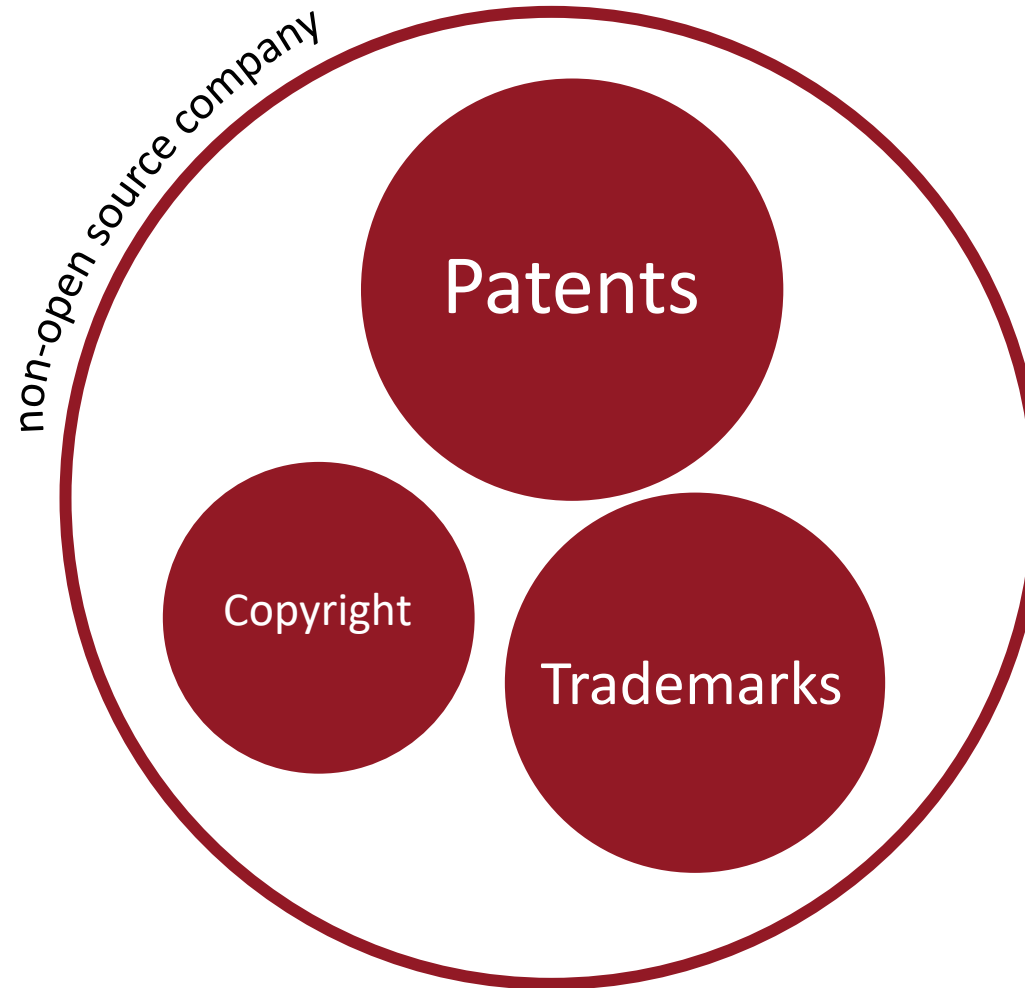
Exclude
third-parties



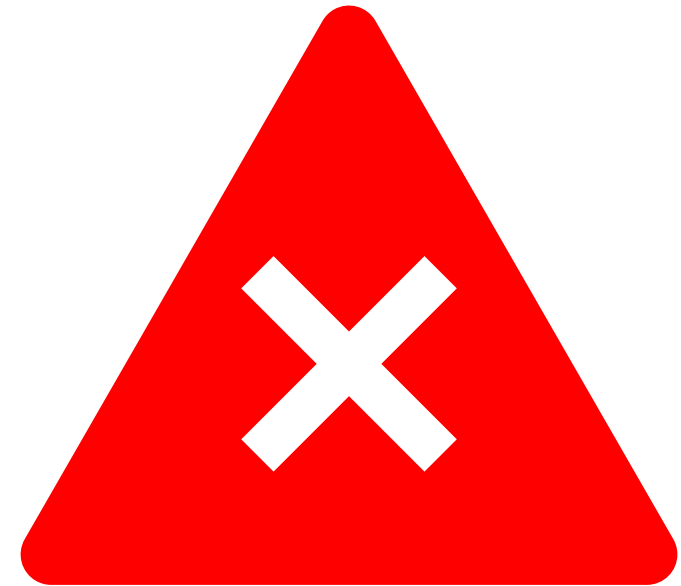
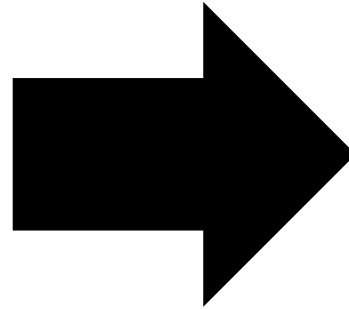
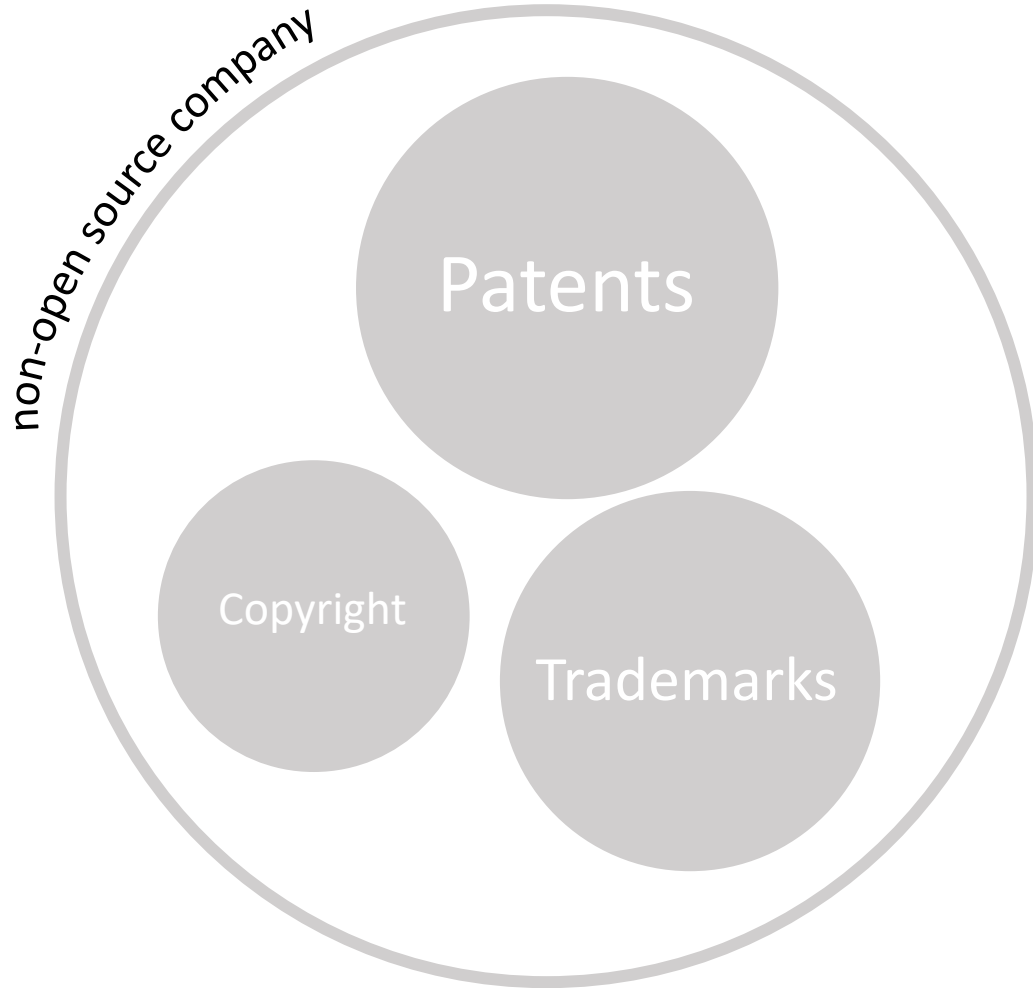
Why are we talking about IP rights?

- Defines the legal framework for openness and enables keeping products and ideas open source;
- Protects creators while encouraging collaboration;
- Supports sustainable business and funding models;
- Builds trust, attribution and community recognition;
- Protects against misuse and proprietary enclosure („Bad Actors“);
- Can pose a risk, especially to smaller projects.

Use cases of IP rights (non-open source)

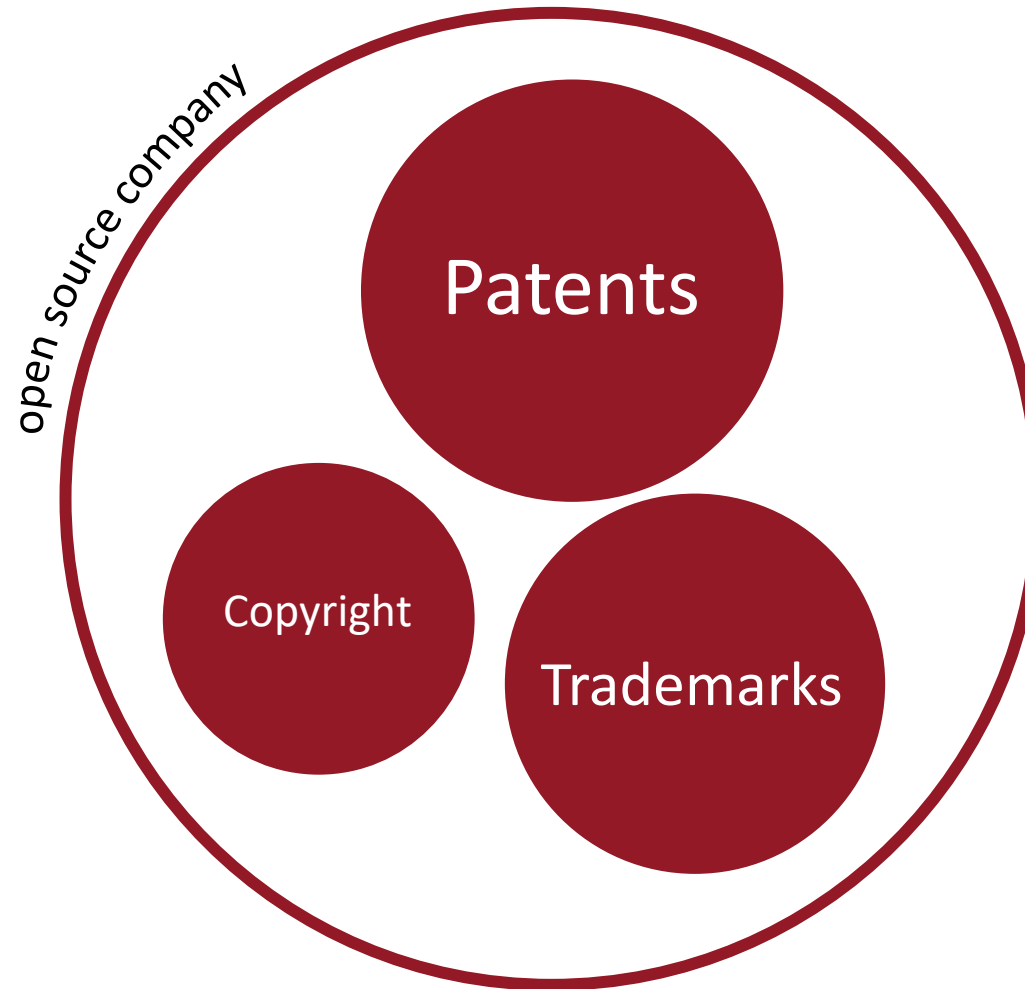


Use cases of IP rights (non-open source)

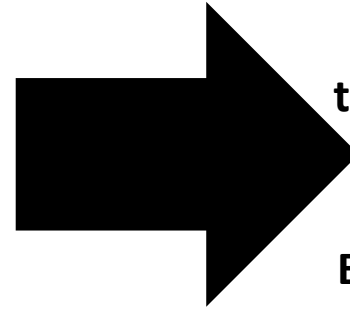
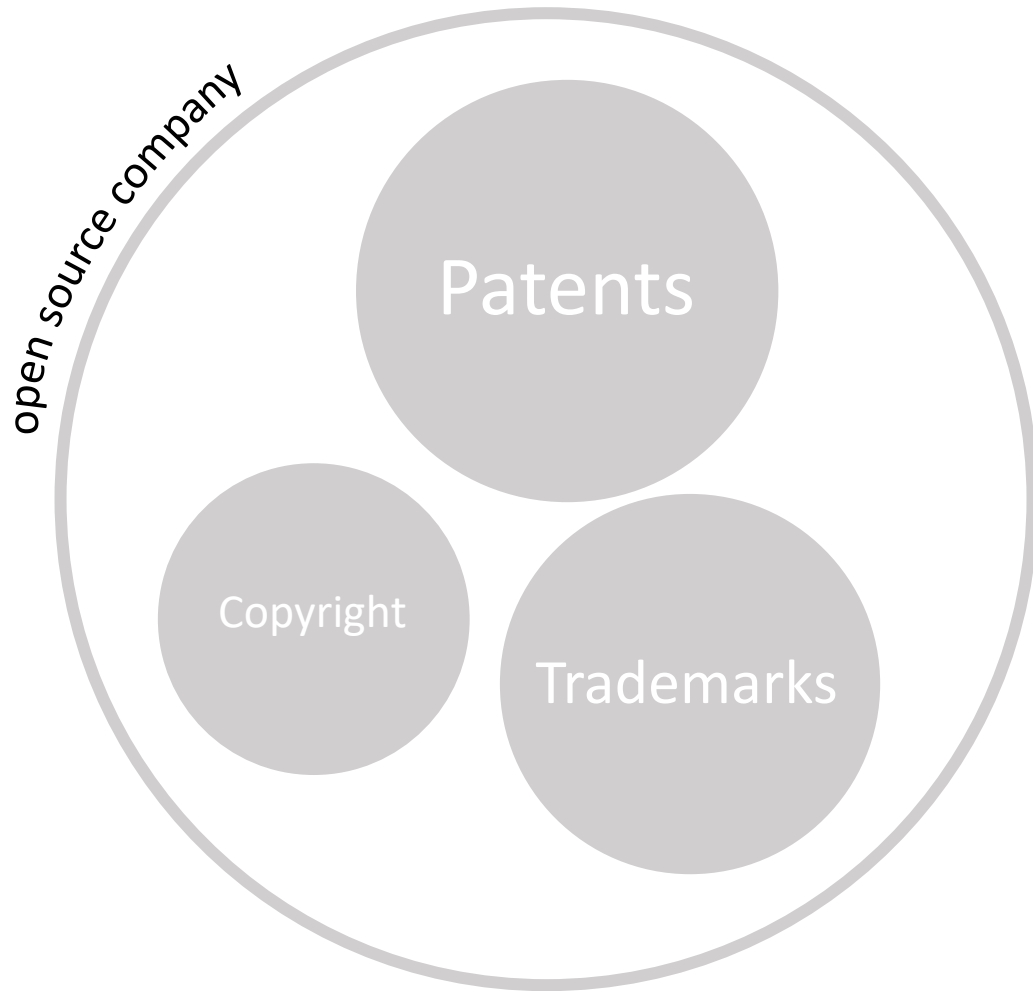


Stops use of similar technology, imagery, documents, etc. by other companies through acquired IP rights.

Use cases of IP rights (open source)



Use cases of IP rights (open source)



Encourages use and improvement of similar technology by acquiring IP rights and making them available under open source terms.

Example: MakerBot was originally one of the most prominent open-source hardware companies in the maker community. After its acquisition by Stratasys in 2013, MakerBot increasingly shifted toward proprietary hardware, software, and licensing models. This led to significant backlash from the open-source community, loss of trust, and criticism over the commercialization of community-driven innovation.

Which IP rights are relevant?

First: Why not simply dedicate everything to the public domain?

- No attribution; no share-alike; risk of enclosure/proprietary; no control over redistribution

Two important questions to ask then:

1. What do you want to protect for how long and where?
2. Which IP right suits the intended protection and your preconditions best?

What do you want to protect for how long and where?

1. Subject matter?

- Software-code or Hardware (singular component or assembled items)
- Logo or Name
- Packaging
- Technical blueprint
- Ideas
- Design of a product

2. Time?

- Some IP Rights expire after a certain amount of time.

3. Location?

- Territoriality

Which IP right suits the intended protection and your preconditions best?

- There is a certain abstract categorization to IP rights and what they protect (idea vs. expression; technical vs. artistic; registered vs. unregistered; multinational vs. national).
- Different costs and maintenance effort.
- Different goals: commercial vs. openness, protecting further innovation or personal identity, business models.
- Each IP right has their own requirements.
- One Open Source Hardware and its components can be protected through different IP rights at the same time.

Open Source Software vs. Open Source Hardware

Open Source Software	Open Source Hardware
Copyright as the central IP right	Multilayered; several IP right might become relevant
Easy and low-cost digital duplication	Besides the distribution of design files, OSH requires physical production, manufacturing, supply chains
Widely harmonized licensing ecosystem	Licensing landscape is more fragmented and less standardized

Which IP right suits the intended protection and your preconditions best?

Let's take a more detailed look at the most relevant IP rights.

2. IP Rights: Patents, Utility Models, Design Rights, Trade Secret, Copyright, Trademarks

Patents

What is a patent?

- Protection of technical inventions (ideas).
- Requires a patentable subject matter, novelty, usefulness/industrial application and an inventive step/non-obviousness.
- Granted for a maximum of 20 years. However, the application process is quite long, detailed and costly.

Registered or unregistered?

- Registered at PTO.

Multinational or national?

- Multinational (e.g. EPÜ), but mostly national (e.g. PatG).

Patents

Why does it matter for OSH?

- Significant risk to your freedom to operate due to the possibility of infringing patents of third parties (cease-and-desist-actions).
- Litigation can be costly and detrimental to the project.
- Patents cover a wide variety of subjects and their claims are not always clear and obvious for a non-lawyer/expert.
- While exposure could be limited at the beginning of the project, omitting these risks can backfire later on. „Defensive publishing“ as a strategy.
- **Category:** Better safe than sorry.
- **Example:** Tesla's Patent Pledge: legally binding commitment made in 2014 where the company irrevocably promised it will not initiate lawsuits against anyone using its technology in good faith.

How to apply for patents:

- Your local PTO will provide detailed information about the application process.

OSH example:

- Patent for a specific hardware component, e.g. motherboard, circuitry, mechanical component.



Utility Models

What is a utility model?

- Can be described as a „small patent“. Protection of technical inventions (ideas).
- Requires a patentable subject matter, novelty, usefulness and an inventive step.
- Faster and cheaper application process than patents, but also less reliable. PTO does not assess application is as much detail which makes utility model vulnerable.
- Granted for a maximum of 10 years.

Registered or unregistered?

- Registered at PTO.

Multinational or national?

- Mostly national (e.g. GebrMG).

Utility Models

Why does it matter for OSH?

- Comparable to patents and therefore the same arguments apply here.
- **Category:** Better safe than sorry.

How to apply for utility models:

- Your local PTO will provide detailed information about the application process.

OSH example:

- Utility model for a specific hardware component, e.g. motherboard, circuitry, mechanical component.

Design Rights

What is a design right?

- Design rights protect the appearance (aesthetic or ornamental features) of products. They do not cover the function of a product. **Rule of thumb:** Patents and utility models protect the inside (functionality) and design rights protect the outside (appearance) of a product.
- The design must be new and original in order to be protected.
- Granted protection for up to 25 years from the date of filing or 3 years for unregistered EU designs.

Registered or unregistered?

- Registered at TMO, but an unregistered EU right is also possible.

Multinational or national?

- Multinational (e.g. European Design Rights) and national (e.g. DesignG).

Design Rights

Why does it matter for OSH?

- Design rights are generally more niche than patents, therefore flying under the radar is more likely.
- However, as with all IP rights, repercussions can be severe, if litigation ensues.
- **Category:** Better safe than sorry.

How to apply for design rights:

- Your local TMO will provide detailed information about the application process.

OSH example:

- Design right for the 3D-appearance of a mechanical component or for the product as a whole.

Trade Secrets

What are trade secrets?

- Trade secrets protect valuable information.
- The information needs to be kept a secret from competitors through verifiable protection measures (burden of proof).
- Trade secrets do not protect against independent discovery of the same information. There is a risk of reverse engineering.
- Granted protection is not limited time-wise.

Registered or unregistered?

- Unregistered.

Multinational or national?

- National (e.g. GeschGehG).

Trade Secrets

Why does it matter for OSH?

- Probably less in comparison to other registered IP rights.
- Yet, trade secrets can become a valuable addition to your own open source strategy (e.g. preparation of IP right application).

How to apply for trade secrets:

- Trade secrets arise naturally and without the need for application once the requirements for protection are met.

OSH example:

- Trade secret in the manufacturing process of a certain hardware component that would otherwise not be protectable or that should be patented later on.

Copyright

What are copyrights?

- Copyrights protect creative works against inter alia reproduction, publication, certain forms of alteration and manipulation, unapproved commercial exploitation.
- A creative work needs to be human-made, perceivable, creative and be characterized by the creator's personality.
- Copyright can cover a wide variety of products, e.g. literature, film, art, software, manuals, logos.
- Copyright protection lasts as long as the lifetime of the authors and a minimum of fifty years after their death. (in Germany: 70 years)

Registered or unregistered?

- Unregistered.

Multinational or national?

- National (e.g. UrhG).

Copyright

Why does it matter for OSH?

- Possible intersection with other IP rights.
- It is difficult to screen every possibility of infringement.
- Legal proceedings can be costly and hinder the development of your project.
- **Rule of thumb:** A basic screening of copyright could be beneficial for most projects.
- Copyright licenses are a great tool to control the openness of your project.

How to apply for copyright:

- Copyrights arise naturally and without the need for application once the requirements for protection are met.

OSH example:

- Copyright for software components or the technical documentation for hardware component or certain images, illustrations, logos, etc.

Trademarks

What are trademarks?

- Trademarks are specific signs (visual, sound, smells, color, etc.).
- They distinguish certain goods and services of one enterprise to those of another = sign of origin.
- Trademarks need to be distinctive in order to be registered and must not be confusingly similar to existing trademarks. The law also states some absolute grounds of refusal (e.g. immorality).
- Certification trademark.
- The granted protection can usually be renewed indefinitely. However, you must use your trademark and be able to prove it.

Registered or unregistered?

- Registered at TMO (e.g. EUIPO or DPMA).

Multinational or national?

- Multinational (e.g. EUTMR) and national (e.g. MarkenG).



Trademarks

Why does it matter for OSH?

- There is a rather sophisticated infrastructure for monitoring trademark infringements, therefore it is more likely that confusingly similar signs do not go unnoticed.
- If trademarks infringements are noticed, legal proceedings can be costly and might hinder the success of your project.
- **Rule of thumb:** A screening of trademarks and subsequent applications could be beneficial for most projects and is usually less costly in comparison to patent rights, etc.
- Trademarks are relatively easy and cheap to obtain and can offer great advantages in managing your project for distribution and marketing.

How to apply for trademarks:

- Your local TMO will provide detailed information about the application process.

OSH example:

- Trademark for the name and logo of your project, e.g. Prusa or Arduino are well known and become trustable through recognition.

Typical Stack in Open Source Hardware

Component/Asset	Possible IP Rights	Example
Firmware/Embedded Software	Copyright	Microcontroller firmware, drivers
CAD files and 3D Models	Copyright, Design Right	STL files for 3D printing
Mechanical components	Patent, Utility Model	Physical OSH, housing structure, technical invention, e.g. sensor system
Product appearance	Design Right, Copyright, Trademark	Shape and visual appearance of devices, packaging
Name and Logo	Trademark, Copyright	Project name, certification logo, community certification
Documentation	Copyright	User manual, assembly guide
Manufacturing Know-how	Trade Secret	Production method, calibration process

3. Licensing – Overview

OSH Licensing - Introduction



- **What is a license?**

- A license is a legal permission granted by a rights holder to another party, allowing them to use certain intellectual property under specified conditions without transferring ownership.

- **Why do we need licenses?**

- They can help to keep OSH open and at the same time maintain control over the use as they provide a clear legal framework.
- They protect the rights of the creator or owner, while enabling access by others.
- They reduce legal uncertainty and risk for both licensors and licensees.

- **What can I license?**

- You can license emergent rights (e.g. copyright) or registered rights (e.g. patents).
- You can limit the scope (e.g. region/country, field, etc.) and duration of the use as well as the circumstances of the reuse (e.g. commercial/non-commercial, sub-licensing, change of license, give attribution, etc.).

OSH Licensing – Need for licenses



FSF Settles Suit Against Cisco

by [Brett Smith](#) — Published on May 20, 2009 10:00 AM

The \$100 Million Court Case for Open Source License Compliance

AYALA GOLDSTEIN, JUNE 1, 2020



OSH Licensing – IP left effect



- **IP right**

- The owner of the IP right has the exclusive right to use, modify and distribute. He can exclude everyone else from the use.

- **Open Source**

- Everyone is allowed to use, modify and (re)distribute.

- **IP left**

- Originally, the term “IP left” derives from the term “copyleft”. It is a play on words intended to illustrate the reverse use of exclusive IP rights in open-source licenses. It exchanges the “right” from “IP right” with the opposite “left”. “Reverse use” means that the rights are not used in the traditional way, i. e. to exclude others, but to make it available to as many people as possible.
- The owner’s exclusive right to copy, publish, distribute and edit a work is used not to exclude, but to allow copying, publishing, distributing and editing by any interested party, as long as they adhere to the corresponding licensing conditions. This usually entails that every contribution to or change of the original source material is also made open-source.
- Types of licenses: **strong IP left** > **weak IP left** > **permissible**

OSH Licensing – Licensing Flavours



○ **Permissive (e.g. CC-BY 4.0 or CERN OHL-P)**

- Everyone can use, modify and (re)distribute.
- No IP left.
- Downstream proprietary use allowed.

○ **Weak IP left (e.g. CERN OHL-W)**

- Everyone can use, modify and (re)distribute.
- IP left applies.
- Only changes to the licensed material itself must be licensed downstream

○ **Strong IP left (e.g. CERN OHL-S; GPLv3)**

- Everyone can use, modify and (re)distribute.
- IP left applies.
- Possibly all material must be licensed downstream.

Lincensor Obligations – General Overview



- Retain existing notices
- Name original author
- Transparency: add modification notice
- Make license text available
- If necessary: observe IP left



Possible strategic Implications

- What is my Design? Which rights apply?
- What is my business model?
 - Selling an OSH-product?
 - Selling related services?
- What is my competitive Edge?
 - OSH Technology itself
 - Personal Know-how
 - Image
 - Lead Time
- What do I want to do with my Design?
 - Make it accessible
 - Control downstream use
 - Commercial or non-profit?
- What should I license?
 - Design, Documentation, Software
 - All or just part?
- Which format and where to license?

OSH Licensing – Pitfalls



- Non-compliance with licence terms.
- Using the wrong license or incompatible licenses (e.g. software and hardware components have different licenses).
- Unclear or missing documentation.
- License overlooks crucial scenarios.
- Lack of license monitoring.

Do's in OSH Licensing



- Attribution
- Transparency
- Complete Licensing
- Make License and Design available
- Check external Material for OSH (or OSS)
- Check licensing scheme (BY-SA-NC-ND?)



Thank You!